Sri Lanka Institute of Information Technology
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Faculty of Computing
Network Design & Management – IT3010
Practical 01: Command Line Network Configuration

Configuring Network Settings on Fedora Client Using CLI

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1. Practical Session Overview

This practical session will include the configuration and validation of two virtual machines' network settings through the use of command line tools. The session will provide an introduction to identifying network interfaces, configuring a static IP address and testing network connectivity in the context of a virtualized environment.A virtual client and a virtual server will be connected via a private network, as well as connected to the internet through the use of Network Address Translation (NAT). Virtualization software will be used to conduct this session; the Linux OS will be used to create a virtualized client and server in order to simulate a client/server network, like those used in the real world.

2. Purpose and Goals of the Experiment

This experiment aims to achieve the following goals:

- Create & configure virtual machines as servers & clients
- Determine network interfaces available from the command line using CLI commands
- Assign a static IP address to the network interfaces
- Enable & verify network connectivity
- Test network connectivity between a server & client machine.

3.Required Tools and System Preparation

- Installation and configuration of a Type-2 hypervisor like VMware Workstation or Oracle VirtualBox on the host operating system.

- Creation of a separate directory tree for virtual machine storage to logically organize and manage the directory.

- Deployment of a CentOS 7 Server virtual machine in a non-graphical mode (command line interface only) to mimic a real production server environment.

- Deployment of Fedora 28 LXDE virtual machine to serve as the client machine with light desktop environments.

- Multiple virtual network interfaces per virtual machine configuration, including;

- VMnet2 is a private host-only network that uses the 10.0.1.0/24 subnet to facilitate communication between VMs.

- NAT adapter configured in a way to give outbound access to outside networks.

- Allocation of proper system resources such as memory, storage, and CPU cores to ensure operational equilibrium.

- Access to the Linux terminal with the superuser (root) to network configuration and service management functions.

- Basic knowledge of Linux networking basics utilities (e.g., ifconfig,ip,nmcli,nmtui) and IPv4 addressing and subnetting and gateway configuration concepts.

4. Configuring Network Settings via Command Line

4.1 Detecting Available Network Interfaces

Command used:

Ifconfig

```
root@mlb-dc1-centos7:~
[root@mlb-dc1-centos7 ~]# ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c2:ce:fe brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.233.128/24 brd 192.168.233.255 scope global dynamic noprefixroute ens33
       valid_lft 1555sec preferred_lft 1555sec
    inet6 fe80::20c:29ff:fec2:cefe/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@mlb-dc1-centos7 ~]#
```

4.2 Reviewing Existing Network Connections

All interfaces were shown here. The NAT adapter displayed a private IP range of 192.168.x.x while also showing the VMnet2 interface as "Not Configured".

ip addr show

```
[root@mlb-dc1-centos7 ~]# ip -4 a s ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    altname enp2s1
    inet 192.168.233.128/24 brd 192.168.233.255 scope global dynamic noprefixroute ens33
       valid_lft 1460sec preferred_lft 1460sec
[root@mlb-dc1-centos7 ~]# hostname -i
fe80::20c:29ff:fec2:cefe%ens33 fe80::87e4:51f0:11c2:ed55%ens36 192.168.233.128 10.0.1.2
[root@mlb-dc1-centos7 ~]# nmcli device show ens33
GENERAL.DEVICE:                         ens33
GENERAL.TYPE:                           ethernet
GENERAL.HWADDR:                         00:0C:29:C2:CE:FE
GENERAL.MTU:                            1500
GENERAL.STATE:                          100 (connected)
GENERAL.CONNECTION:                     NAT
GENERAL.CON-PATH:                       /org/freedesktop/NetworkManager/ActiveConnection/4
WIRED-PROPERTIES.CARRIER:               on
IP4.ADDRESS[1]:                         192.168.233.128/24
IP4.GATEWAY:                            192.168.233.2
IP4.ROUTE[1]:                           dst = 192.168.233.0/24, nh = 0.0.0.0, mt = 102
IP4.ROUTE[2]:                           dst = 0.0.0.0/0, nh = 192.168.233.2, mt = 102
IP4.DNS[1]:                             192.168.233.2
IP4.DOMAIN[1]:                          localdomain
IP6.ADDRESS[1]:                         fe80::20c:29ff:fec2:cefe/64
IP6.GATEWAY:                            --
IP6.ROUTE[1]:                           dst = fe80::/64, nh = ::, mt = 1024
[root@mlb-dc1-centos7 ~]#
```

This helped determine what all interfaces currently have IP addresses and if the VMnet2 interface was inactive.

4.3 Analyzing Network Interface Information

Command used:

nmcli device status

This provides a means to see all available physical and virtual interfaces and their status.

```
[root@mlb-dc1-centos7 ~]# nmcli connection up ens33
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
[root@mlb-dc1-centos7 ~]#
```

4.4 Setting a Manual (Static) IP Address

On Fedora client:

nmcli con add type ethernet con-name vmnet2 ifname ens160 ip4 10.0.1.3/24 gw4 10.0.1.1

On CentOS server (via nmtui):

- IP: 10.0.1.2/24
- Gateway: 10.0.1.1

```
root@ndm-cli-fedora28:/home/client1
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.addresses 10.0.1.3/24
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.gateway 10.0.1.1
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.method manual
root@ndm-cli-fedora28:/home/client1# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
root@ndm-cli-fedora28:/home/client1#
```

4.5 Initiating the Network Interface

Command used:

nmcli con up ens160

```
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.addresses 10.0.1.3/24
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1#
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.gateway 10.0.1.1
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.method manual
root@ndm-cli-fedora28:/home/client1# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/6)
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.gateway 10.0.1.1
root@ndm-cli-fedora28:/home/client1# nmcli connection modify ens160 ipv4.method manual
root@ndm-cli-fedora28:/home/client1# nmcli connection up ens160
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
root@ndm-cli-fedora28:/home/client1# ip addr show ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:46:5f:22 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::2b32:f825:1f10:13e6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@ndm-cli-fedora28:/home/client1#
```

This activated the connection.

5. Network Testing and Confirmation

5.1 Validating IP Configuration

Command used:

ip addr show ens160

```
root@ndm-cli-fedora28:/home/client1# ip addr show ens160
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:46:5f:22 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::2b32:f825:1f10:13e6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Confirmed correct IP assignment.

5.2 Displaying All Active and Inactive Interfaces

Command used:
Ip addr show
i

```
root@ndm-cli-fedora28:/home/client1# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:46:5f:22 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
       valid_lft forever preferred_lft forever
    inet6 fe80::2b32:f825:1f10:13e6/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:46:5f:2c brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens224
       valid_lft forever preferred_lft forever
    inet6 fe80::197c:7397:4574:2bac/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@ndm-cli-fedora28:/home/client1#
```

This command provides detailed interface information.

5.3 Monitoring Connection and Link Status

Command used:

nmcli device status

```
root@ndm-cli-fedora28:/home/client1# nmcli device status
DEVICE  TYPE      STATE                   CONNECTION
ens224  ethernet  connected               ens224
ens160  ethernet  connected               ens160
lo      loopback  connected (externally)  lo
root@ndm-cli-fedora28:/home/client1#
```

This verified that the necessary interfaces were operational and connected.

5.4 Verifying Communication Between Client and Server Systems

Command used:

ping 10.0.1.2

```
root@ndm-cli-fedora28:/home/client1# ping 10.0.1.2 -c 4
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=3.34 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=0.807 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=1.56 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=2.10 ms

--- 10.0.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3040ms
rtt min/avg/max/mdev = 0.807/1.951/3.341/0.923 ms
root@ndm-cli-fedora28:/home/client1#
```

Successful response confirmed connectivity between client and server.

5.5 Reviewing Advanced Network Details

As a final optional verification, the complete connection details can be reviewed:

nmcli connection show ens160

```
[root@ndm-cli-fedora28 ~]# nmcli connection show ens160
connection.id:                          ens160
connection.uuid:                        d094ca58-0fef-372b-afed-24552652c90c
connection.stable-id:                   --
connection.type:                        802-3-ethernet
connection.interface-name:              ens160
connection.autoconnect:                 yes
connection.autoconnect-priority:        -999
connection.autoconnect-retries:         -1 (default)
connection.auth-retries:                -1
connection.timestamp:                   1771137556
connection.read-only:                   no
connection.permissions:                 --
connection.zone:                        --
connection.master:                      --
connection.slave-type:                  --
connection.autoconnect-slaves:          -1 (default)
connection.secondaries:                 --
connection.gateway-ping-timeout:        0
connection.metered:                     unknown
connection.lldp:                        default
802-3-ethernet.port:                    --
802-3-ethernet.speed:                   0
802-3-ethernet.duplex:                  --
802-3-ethernet.auto-negotiate:          no
802-3-ethernet.mac-address:             --
802-3-ethernet.cloned-mac-address:      --
802-3-ethernet.generate-mac-address-mask:--
802-3-ethernet.mac-address-blacklist:   --
802-3-ethernet.mtu:                     auto
802-3-ethernet.s390-subchannels:        --
802-3-ethernet.s390-nettype:            --
802-3-ethernet.s390-options:            --
802-3-ethernet.wake-on-lan:             default
802-3-ethernet.wake-on-lan-password:    --
ipv4.method:                            manual
ipv4.dns:                               --
ipv4.dns-search:                        --
ipv4.dns-options:                       ""
ipv4.dns-priority:                      0
ipv4.addresses:                         10.0.1.3/24
ipv4.gateway:                           10.0.1.1
ipv4.routes:                            --
ipv4.route-metric:                      -1
ipv4.route-table:                       0 (unspec)
ipv4.ignore-auto-routes:                no
ipv4.ignore-auto-dns:                   no
lines 1-46
```

6. Problem Resolution and Diagnostic Procedures

6.1 Frequently Encountered Issues and Fixes

- Problem Number 1 - The Interface doesn't Activate
- Solution - Restart the NetworkManager service: systemctl restart NetworkManager.
- Problem Number 2 - Incorrectly assigned IP Address
- Solution - Reconfigure the connection either using nmcli or nmtui: nmcli con mod or nmtui.
- Problem Number 3 - There is no Connectivity
- Solution - Check for Firewall issues. Also, make sure that both VM(s) are on the same subnet (VMnet2).

7.Final Observations and Conclusions

Using virtualization, practical project showcased an operationally viable configuration of the static IP address for communicating with both a grow workstation (Fedora client) and a services workstation (CentOS server). It also included command line interface (CLI) based configuration to help gain a better understanding of managing networks and troubleshooting network issues.

Appendix A: Command-Line Instructions Overview

Command	Description
nmcli connection show ens160	Show details of the ens160 connection
ping 10.0.1.2	Test connectivity to IP 10.0.1.2
nmcli device status	Display status of all network interfaces
ip addr show	Show IP addresses of all interfaces
ip addr show ens160	Show IP address of ens160 interface
nmcli con up ens160	Bring up the ens160 connection
nmcli con add type ethernet con-name vmnet2 ifname <interface_name> ip4 10.0.1.3/24 gw4 10.0.1.1	Add new Ethernet connection vmnet2 with IP and gateway
nmcli device status	Verify status of network devices
ip addr show	Check assigned IP addresses
ifconfig	Legacy command to display interfaces and IP info